

Parent Functions I: $\sqrt{x}$ and $|x|$

Algebra II | Unit 1 | Day 4 | TEK A2.2.A

Objective: I can graph the square root and absolute value parent functions from a table and analyze their key attributes: domain, range, intercepts, symmetry, and minimum.

Vocabulary

- Parent function** — the _____ function in a family, with nothing added, subtracted, or multiplied.
- x-intercept** — where the graph crosses the x-axis. There, $y = \underline{\hspace{1cm}}$. **y-intercept** — there, $x = \underline{\hspace{1cm}}$.
- Symmetry** — the graph can be _____ onto itself.
- Minimum** — the _____ y-value the function reaches.

Part 1 — The Square Root Function, $f(x) = \sqrt{x}$

Fill in the table. Watch what happens with negative inputs.

x	-4	-1	0	1	4	9
$f(x) = \sqrt{x}$						

Why are the first two undefined? _____

Part 2 — The Absolute Value Function, $f(x) = |x|$

Absolute value gives distance from zero, so the output is never negative.

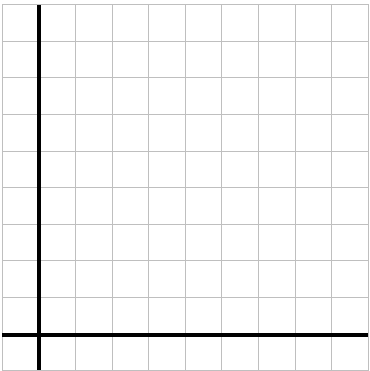
x	-5	-2	-1	0	1	2	5
$f(x) = x $							

What shape does this make? _____

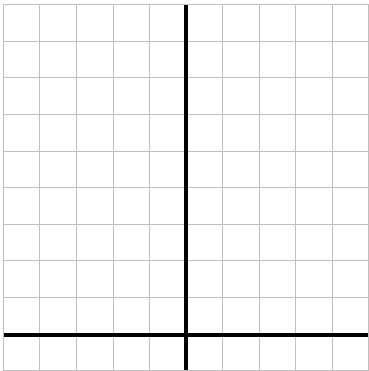
Part 3 — Graph Them Both

Each square is 1 unit. Plot your table values, then connect them smoothly.

A. $f(x) = \sqrt{x}$ (x from 0 to 9)



B. $f(x) = |x|$ (x from -5 to 5)



Which quadrant only? _____

Fold line? _____

Part 4 — Compare the Key Attributes

Attribute	$f(x) = \sqrt{x}$	$f(x) = x $
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Domain (interval)	_____	_____
Range (interval)	_____	_____
x- and y-intercept	_____	_____
Symmetry	_____	_____
Minimum	_____	_____
Increasing / decreasing	_____	_____

Biggest difference: their _____. Only _____ refuses negative inputs.

Part 5 — Name That Function

Use only the attributes given. No equation.

- Domain and range both $[0, \infty)$, no symmetry, always increasing. _____
- Domain all reals, range $[0, \infty)$, symmetric across the y-axis, minimum 0. _____

Practice

Evaluate, then answer. Show your work where there is room.

- Evaluate: $\sqrt{25} =$ _____ $\sqrt{0} =$ _____ $|-7| =$ _____ $|0| =$ _____

- Explain why $\sqrt{-9}$ is not a real number.

- Name every x-value where $|x| = 4$. _____ How many? _____

- Name every x-value where $\sqrt{x} = 4$. _____ How many? _____

- Items 3 and 4 have different numbers of answers. What does that tell you about the two graphs?

- On $[1, 4]$, give the minimum and maximum of $f(x) = \sqrt{x}$. min _____ max _____

Exit Ticket

Both functions have a minimum of 0 and pass through the origin. Give one attribute that tells them apart, and explain why it happens.

Attribute: _____

Why: _____